

Alejandro Jaimes

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EDUCATION

University of Central Florida

Bachelor of Science in Computer Science

Orlando, FL

May 2028

EXPERIENCE

Software Engineer Intern

May 2026 - Present

Institute for Simulation and Training

Orlando, FL

- Constructed a real-time spatial anchoring system using **AR Foundation**, enabling **8** users within **5cm** accuracy.
- Developed networked player-tracker with **Photon**, matching data for **XR** devices at low latency for live demos.
- Implemented runtime **XR subsystem** detection for hardware-aware kiosk mode, reducing demo setup by **100%**.
- Designed backward-compatible multiplayer integration system, maintaining **100%** single-player functionality while adding new multiplayer features across **2** distinct codepaths.

President

Apr 2026 - Present

Graphics Programming Knights

Orlando, FL

- Established Graphics Programming Knights, a non-profit student organization, growing membership to **165+**.
- Organized the first community-wide Render Jam competition resulting in **40+** participants and **5** projects.
- Orchestrated **2 workshops per term** on pathways into graphics programming, covering essential **mathematics**, **game development engines**, and **GPU programming** with student attendance of **20+**.
- Expanded club reach by **40%**, establishing partnerships to have workshops and hackathons with **3** organizations.

Outreach Board

Jan 2026 - Present

Knight Hacks

Orlando, FL

- Boosted online community engagement online with strategic content creation and event awareness by **57%**.
- Produced and edited Reels, increasing Knight Hacks' visibility and follower engagement, averaging **5k+** views.
- Expanded event attendance by **15%** compared to previous semesters through targeted outreach campaigns on social media and speaking to the public, increasing participation with an average of **40+** people per meeting.

PROJECTS

greghouse | *C++, Arduino, ESP32, React, AWS, Node.js, Lambda, DynamoDB*

- **3rd place winner** of the **Best Overall** category at BloomKnights out of **86** projects and **240+** participants.
- Engineered **ESP32** firmware in **C++** streaming soil telemetry to **AWS** every 2s with sub-3s latency.
- Architected an IoT pipeline with **API Gateway**, **Lambda**, and **DynamoDB** handling **1,800+** readings hourly.
- Crafted a pixel-art **TFT** display on **C++** with mood-driven animations, synchronizing with the **React** app.

sFun (SNES emulator) | *C++, C, CMake, Assembly, SDL2*

- Designed and implemented a **65c816 CPU** emulator supporting, dynamic register width handling using **C++**.
- Built an extensible opcode dispatch system using function pointer tables and implemented core CPU instructions (LDA, REP/SEP, branching, stack operations, JSR/RTS) with correct flag behavior.
- Developed a memory subsystem for SNES RAM/ROM access, enabling instruction fetch, decoding, and execution.
- Engineered robust control flow correctness for relative branching, subroutine calls, and stack-based return handling ensuring **100%** correctness on test cases.

Next Step | *React, Tailwind, Next.js, TypeScript, Python, Flask, Vapi*

- **3rd place winner** of State Farm's Sponsor Challenge at Shell Hacks 2025 out of **50** projects submitted.
- Created an AI voice agent delivering interactive, personalized feedback across **3** different insurance categories.
- Implemented backend services to process and structure conversational data generating dynamic agent responses, reducing response time by **66%** using **Python**, integrated with the **Vapi API**.
- Engineered **7** interactive components for optimal experience with **TypeScript**, **Next.js**, and **React** for frontend.

TECHNICAL SKILLS

Languages: C, C++, C#, Python, HTML, CSS, JavaScript, TypeScript, Assembly, Java

Frameworks/Libraries: Unity, Next.js, Node.js, Tailwind CSS, React, Flask, Unreal Engine, Three.js, OpenCV, Mediapipe, Vapi, SLD2, OpenGL, CUDA, CMake